

Homework 1 Report

Section 0 - Personal Configuration and Domain

Value	Definition
SID4	2383
PORT_BASE	8583
PREFIX	s2383
SEED	2383
VERIFY_SEED	262383
DOMAIN_ID	7
Hardware	MacBook Pro, Apple M4 (10-core CPU), 16 GB RAM
Local model	qwen3:8b through Ollama
Tagged commit hash	

Part 1

I. HTML (Note: the output screenshot is part of the DEPLOYMENT section)

```
<!DOCTYPE html>
<html>

<head>
  <title>HW1-Ananya Yallapragada</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      max-width: 600px;
      margin: 40px auto;
      padding: 20px;
      background-color: #badde9;
    }

    h1 {
      text-align: center;
      color: #333;
    }

    form {
      background-color: rgb(255, 255, 255);
      padding: 25px;
      border-radius: 8px;
      box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
    }

    label {
      font-weight: bold;
    }

    input[type="text"],
    input[type="email"],
    input[type="date"],
    input[type="time"],
    select,
    textarea {
      width: 100%;
      padding: 8px;
      margin: 5px 0 15px;
```

```

        box-sizing: border-box;
    }

    fieldset {
        margin-bottom: 15px;
    }

    input[type="submit"] {
        background-color: #4CAF50;
        color: white;
        padding: 10px 20px;
        border: none;
        cursor: pointer;
    }
</style>
</head>

<body>

    <h1>Community sports league fixtures</h1>
    <center>
        <form id="fixtureForm">
            <label for="fixtureName">Fixture Name:</label><br>
            <input type="text" id="fixtureName" name="fixtureName" value=""
                placeholder="e.g., Round 1, Semifinal, or Final" required
size="30"><br>
            <fieldset> <!-- grouping related elements -->
                <legend>Teams</legend>
                <label for="team1">Team 1:</label><br>
                <input type="text" id="team1" name="team1" placeholder="Enter
the first team name" required
                size="30"><br>
                <label for="team2">Team 2:</label><br>
                <input type="text" id="team2" name="team2" placeholder="Enter
the second team name" required
                size="30"><br>
            </fieldset>
            <label for="date">Date:</label><br>
            <input type="date" id="date" name="date" value="" required><br>
            <label for="time">Time:</label><br>
            <input type="time" id="time" name="time" value="" required><br>
            <label for="location">Location:</label><br>

```

```
<input type="text" id="location" name="location" value=""
placeholder="eg. Spartan Stadium" required
size="30"><br>
<label for="submitterEmail">Submitter Email:</label><br>
<input type="email" id="submitterEmail" name="submitterEmail"
value=""
placeholder="e.g. leaguemanager@email.com" required
size="30"><br>
<label for="description">Description:</label><br>
<textarea id="description" name="description" rows="5" cols="50"
placeholder="Enter additional fixture details, such as arrival
instructions, match rules, or schedule notes"
required></textarea><br>
<label for="sportType">Sport Type:</label><br>
<select name="sportType" id="sportType">
  <option value="">Select Sport</option>
  <option value="soccer">Soccer</option>
  <option value="basketball">Basketball</option>
  <option value="tennis">Tennis</option>
  <option value="pickleball">Pickleball</option>
</select><br>
<input type="checkbox" id="termsAccepted" name="termsAccepted"
value="">
<label for="termsAccepted">I agree to the terms and
conditions.</label><br>
<input type="submit" value="Create Fixture">
</form>
</center>
<script src="script.js"></script>
</body>
</html>
```

II. Javascript (Note: the output screenshot is part of the DEPLOYMENT section)

```
const form = document.getElementById('fixtureForm');
const content = document.getElementById('description');
const checkbox = document.getElementById('termsAccepted');

const closureCounterSubmission = function () {
  let count = 0;
  return function () {
    count++;
    return count;
  }
};

const submissionCount = closureCounterSubmission();

const validateForm = (event) => {
  // Question 1
  event.preventDefault(); // to stop the default submission of the form
  if (content.value.trim().length <= 25) {
    alert("Description must be at least 26 characters long.");
    return;
  }
  if (!checkbox.checked) {
    alert("You must accept the terms and conditions.");
    return;
  }

  // Question 2
  const formData = new FormData(form); // collect values from the form
  const formObject = Object.fromEntries(formData.entries());
  formObject.termsAccepted = checkbox.checked; // because if unchecked,
  // formData does not include the key termsAccepted
  const jsonData = JSON.stringify(formObject); // convert object to json
  console.log(jsonData);

  // Question 3
  const parseObject = JSON.parse(jsonData); // convert json back to object
  const { fixtureName, submitterEmail } = parseObject;
  console.log('Fixture name:', fixtureName);
  console.log('Email id', submitterEmail);

  // Question 4
  const updatedObject = {
```

```

        ...parseObject,
        submissionDate: new Date().toISOString(), // ISO string represents
current date-time
    }
    console.log("Submission date updated:", updatedObject.submissionDate)

    // Question 5

    console.log("Form submissions count:", submissionCount());
};

form.addEventListener("submit", validateForm); // attaching the arrow function
to the form's submit event

```

The screenshot displays a web browser window with a form titled "Community sports league fixtures" and the Chrome DevTools console open to the right.

Form Details:

- Fixture Name:** Round 1
- Teams:**
 - Team 1:** Tiger
 - Team 2:** Panthers
- Date:** 08/31/2026
- Time:** 05:30 PM
- Location:** Spartan Stadium
- Submitter Email:** leaguemanager@gmail.com
- Description:** Teams must arrive 30 mins before match time. Snacks will be provided during half-time.
- Sport Type:** Basketball
- ☒ I agree to the terms and conditions.
- Create Fixture** (button)

DevTools Console Output:

```

Failed to load resources: the server responded with a status of 404 (File not found)
[{"fixtureName":"Round 1","team1":"Tiger","team2":"Panthers","date":"2026-08-31","time":"17:30","location":"Spartan Stadium","submitterEmail":"leaguemanager@gmail.com","description":"Teams must arrive 30 mins before match time. Snacks will be provided during half-time.","sportType":"basketball","termsAccepted":true}]
Fixture name: Round 1
Email id leaguemanager@gmail.com
Submission date updated:
{"fixtureName": "Round 1", team1: "Tiger", team2: "Panthers", date: "2026-08-31", time: "17:30", ...}
  description: "Teams must arrive 30 mins before match time. Snacks will be provided during half-time. "
  fixtureName: "Round 1"
  location: "Spartan Stadium"
  sportType: "basketball"
  submissionDate: "2026-08-31T20:21:45.988Z"
  submitterEmail: "leaguemanager@gmail.com"
  team1: "Tiger"
  team2: "Panthers"
  termsAccepted: true
  time: "17:30"
  [[Prototype]]: Object
Form submissions count: 1

```

DEPLOYMENT: (NOTE: consists of the output of the HTML and Javascript sections - the localhost page)

Docker:

localhost:8583

Community sports league fixtures

Fixture Name:
e.g., Round 1, Semifinal, or Final

Teams

Team 1:
Enter the first team name

Team 2:
Enter the second team name

Date:
mm/dd/yyyy

Time:
--:--:--

Location:
eg. Spartan Stadium

Submitter Email:
e.g. leaguemanager@email.com

Description:
Enter additional fixture details, such as arrival instructions, match rules, or schedule notes

Sport Type:
Select Sport

☐ I agree to the terms and conditions.

Create Fixture

data260-hw1 latest 2c5309b158e6 2 minutes ago 62.29 MB

Containers [Give feedback](#)

Container CPU usage 0.00% / 1000% (10 CPUs available)

Container memory usage 10.1MB / 7.57GB [Show charts](#)

Search

Only show running containers

	Name	Container ID	Image	Port(s)	CPU (%)	Memory usage	Memory (%)	Disk read/write	Network I/O	PIDS	Last started	Actions
☐	data260-hw1-container	2cabf3898381	data260-hw1	8583-80	0%	10.1MB / 7.57GB	0.13%	0B / 8.19KB	3.92KB / 7.49KB	11	28 seconds ago	🔄 ⚙️ 🗑️

AWS ECS:

Community sports league fixtures

Fixture Name:
e.g., Round 1, Semifinal, or Final

Teams

Team 1:
Enter the first team name

Team 2:
Enter the second team name

Date:
mm/dd/yyyy

Time:
--:-- --

Location:
eg. Spartan Stadium

Submitter Email:
e.g. leaguemanager@email.com

Description:
Enter additional fixture details, such as arrival instructions, match rules, or schedule notes

Sport Type:
Select Sport

☐ I agree to the terms and conditions.

Create Fixture

Tasks (1)

Last updated August 30, 2026, 22:04 (UTC-7:00)

Manage tags Stop Run new task

Filter tasks by property or value

Filter desired status Any desired status

Filter launch type Any launch type

☐	Task	Last status	Desired status	Task definition	Health status	Started by
☐	e75c7ac0e6494004b...	Running	Running	data260-task:1	Unknown	ecs-svc/79834084...

Services (1) Info

Last updated August 30, 2026, 22:04 (UTC-7:00)

Manage tags Update Delete service Create

Filter services by value

Filter launch type Any launch type

Filter scheduling strategy Any scheduling strategy

Filter resource management type Any resource management type

☐	Service name	ARN	Status	Schedu...	L...	Task de...	Deployments and tasks
☐	data260-task-service	arn:aws:ec...	Active	Replica	Far...	data260-t...	1/1 Tasks

Part 2

Agentic AI

1. **Command Used** - `python code/agents_demo.py \ --title "Community Soccer Semifinal" \ --content "The Falcons and Tigers will compete at Spartan Stadium on September 12. Both teams must arrive 30 minutes early for player check-in." \ --model qwen3:8b \ --strict`

```
1 python code/agents_demo.py \
2 --title "Community Soccer Semifinal" \
3 --content "The Falcons and Tigers will compete at Spartan Stadium on September 12. Both teams must arrive 30 minutes early for player check-in." \
4 --model qwen3:8b \
5 --strict
```

2. **Console Screenshots**

```
--- Planner (143393 ms) ---
{
  "thought": "Identifying key elements: teams, venue, date, and check-in requirements.",
  "message": "The Falcons and Tigers will compete in the Community Soccer Semifinal at Spartan Stadium on September 12, requiring both teams to arrive 30 minutes early for check-in.",
  "data": {
    "tags": [
      "community soccer match",
      "spartan stadium game",
      "team check-in requirements"
    ],
    "summary": "Falcons and Tigers face off in a Community Soccer Semifinal at Spartan Stadium on September 12 with early arrival needed.",
    "issues": []
  }
}
```

```
--- Reviewer (33535 ms) ---
{
  "thought": "Tags focus on event specifics, venue, and logistics. Summary concisely captures key details within word limit.",
  "message": "The Falcons and Tigers will compete in the Community Soccer Semifinal at Spartan Stadium on September 12, requiring both teams to arrive 30 minutes early for check-in.",
  "data": {
    "tags": [
      "community soccer semifinal",
      "spartan stadium event",
      "team check-in protocol"
    ],
    "summary": "Falcons and Tigers play Community Soccer Semifinal at Spartan Stadium on September 12 with early arrival required.",
    "issues": []
  }
}
```

```

Finalized Output
{
  "thought": "Finalizing tags to reflect event type, venue, and check-in logistics. Summary remains concise and accurate.",
  "message": "Falcons and Tigers compete in Community Soccer Semifinal at Spartan Stadium on September 12 with early check-in required.",
  "data": {
    "tags": [
      "community soccer semifinal",
      "spartan stadium event",
      "team check-in protocol"
    ],
    "summary": "Falcons and Tigers play Community Soccer Semifinal at Spartan Stadium on September 12 with early arrival required.",
    "issues": []
  }
}

```

```

Publish Package
{
  "title": "Community Soccer Semifinal",
  "email": "student@example.com",
  "content": "The Falcons and Tigers will compete at Spartan Stadium on September 12. Both teams must arrive 30 minutes early for player check-in.",
  "agents": {
    "transcript": [
      {
        "role": "Planner",
        "content": "{ \"thought\": \"Identifying key elements: teams, venue, date, and check-in requirements.\", \"message\": \"The Falcons and Tigers will compete in the Community Soccer Semifinal at Spartan Stadium on September 12, requiring both teams to arrive 30 minutes early for check-in.\", \"data\": { \"tags\": [\"community soccer match\", \"spartan stadium game\", \"team check-in requirements\"], \"summary\": \"Falcons and Tigers face off in a Community Soccer Semifinal at Spartan Stadium on September 12 with early arrival needed.\", \"issues\": [] } }"
      },
      {
        "role": "Reviewer",
        "content": "{ \"thought\": \"Tags focus on event specifics, venue, and logistics. Summary concisely captures key details within word limit.\", \"message\": \"The Falcons and Tigers will compete in the Community Soccer Semifinal at Spartan Stadium on September 12, requiring both teams to arrive 30 minutes early for check-in.\", \"data\": { \"tags\": [\"community soccer semifinal\", \"spartan stadium event\", \"team check-in protocol\"], \"summary\": \"Falcons and Tigers play Community Soccer Semifinal at Spartan Stadium on September 12 with early arrival required.\", \"issues\": [] } }"
      }
    ],
    "final": {
      "tags": [
        "community soccer semifinal",
        "spartan stadium event",
        "team check-in protocol"
      ],
      "summary": "Falcons and Tigers play Community Soccer Semifinal at Spartan Stadium on September 12 with early arrival required.",
      "issues": []
    }
  },
  "submissionDate": "2026-08-31T08:12:46Z"
}

```

3. Short Answers -

- 3 final tags** - "community soccer semifinal", "spartan stadium event", "team check-in protocol"
- The final summary** - Falcons and Tigers play Community Soccer Semifinal at Spartan Stadium on September 12 with early arrival required.
- Reviewer** - Yes. The Reviewer refined the Planner's tags from broader phrases such as "community soccer match" to more specific phrases such as "community

soccer semifinal.” It also shortened the summary while preserving the essential fixture details.

d. Steps -

- i. The Planner examines the supplied title and content, identifies the principal event details, and proposes three tags and a summary in the required JSON structure.
- ii. The Reviewer receives the Planner’s response and the original input. It checks the relevance of the tags, summary accuracy, output structure, and word limit, and then proposes corrections where necessary.
- iii. The finalization step selects and normalizes the reviewed result. It enforces exactly three tags, a summary of no more than 25 words, and valid JSON.
- iv. The publish package combines the original title and content with the agent transcript, final tags, final summary, issues, and submission timestamp.

Part 3

Measuring Non-Determinism

Upon observing the table in METRICS.md, we can draw some inferences. For example, if two users send the identical fixture update, they can receive different tags. At temperature 0.0, the output was relatively stable: all 20 runs included 'soccer match rescheduling', and only two distinct three-tag sets occurred. At temperature 0.7, the output was much less stable: all 20 runs produced different three-tag sets, and no exact tag appeared in every run. The higher-temperature runs were also slower in this experiment: p50 latency was 103,358 ms at temperature 0.7, compared with 82,441 ms at temperature 0.0.

Variation per run is acceptable when generating optional discovery labels for a community-sports post, because several relevant tag phrasings can help users browse the same event. It is not acceptable when the output drives an operational action, such as assigning a fixture to the correct venue or notifying teams of a schedule change, because inconsistent wording could route or communicate the update incorrectly.

```
• (base) annie@mac data260-2383 % python code/run_nondeterminism.py --help
2026-08-31T09:07:16.712289+00:00 Starting temperature=0.7, run=1
Completed temperature=0.7, run=1, latency_ms=115146, tags=['soccer game rescheduling', 'venue change announcement', 'logistical updates']
2026-08-31T09:09:11.860669+00:00 Starting temperature=0.7, run=2
Completed temperature=0.7, run=2, latency_ms=119965, tags=['soccer match rescheduling', 'venue change notice', 'logistics update']
2026-08-31T09:11:11.828771+00:00 Starting temperature=0.7, run=3
Completed temperature=0.7, run=3, latency_ms=134565, tags=['soccer game rescheduling', 'venue change notification', 'logistics update']
2026-08-31T09:13:26.396019+00:00 Starting temperature=0.7, run=4
Completed temperature=0.7, run=4, latency_ms=142509, tags=['soccer match rescheduling', 'venue change notification', 'event logistics update']
2026-08-31T09:15:48.908012+00:00 Starting temperature=0.7, run=5
Completed temperature=0.7, run=5, latency_ms=113449, tags=['soccer game rescheduling', 'venue change notification', 'event logistics update']
2026-08-31T09:17:42.358746+00:00 Starting temperature=0.7, run=6
Completed temperature=0.7, run=6, latency_ms=112268, tags=['soccer semifinal reschedule', 'venue change notification', 'rain backup venue info']
2026-08-31T09:19:34.628286+00:00 Starting temperature=0.7, run=7
Completed temperature=0.7, run=7, latency_ms=91446, tags=['soccer match rescheduling', 'stadium venue change', 'event logistics update']
2026-08-31T09:21:06.075683+00:00 Starting temperature=0.7, run=8
Completed temperature=0.7, run=8, latency_ms=103940, tags=['community soccer semifinal rescheduling', 'spartan stadium location change', 'logistics and backup venue']
2026-08-31T09:22:50.016816+00:00 Starting temperature=0.7, run=9
Completed temperature=0.7, run=9, latency_ms=84814, tags=['soccer semifinal rescheduling', 'venue change notification', 'community event logistics']
2026-08-31T09:24:14.831636+00:00 Starting temperature=0.7, run=10
Completed temperature=0.7, run=10, latency_ms=79472, tags=['soccer semifinal rescheduling', 'field maintenance update', 'event logistics changes']
2026-08-31T09:25:34.304613+00:00 Starting temperature=0.7, run=11
Completed temperature=0.7, run=11, latency_ms=83270, tags=['soccer match rescheduling', 'venue change announcement', 'event logistics update']
2026-08-31T09:26:57.600226+00:00 Starting temperature=0.7, run=12
Completed temperature=0.7, run=12, latency_ms=84928, tags=['soccer game rescheduling', 'field maintenance impact', 'venue change logistics']
2026-08-31T09:28:22.538409+00:00 Starting temperature=0.7, run=13
Completed temperature=0.7, run=13, latency_ms=83735, tags=['soccer match rescheduling', 'field maintenance impact', 'backup venue preparation']
2026-08-31T09:29:46.276489+00:00 Starting temperature=0.7, run=14
Completed temperature=0.7, run=14, latency_ms=102776, tags=['soccer semifinal rescheduling', 'field maintenance notice', 'venue logistics update']
2026-08-31T09:31:29.056431+00:00 Starting temperature=0.7, run=15
Completed temperature=0.7, run=15, latency_ms=74064, tags=['soccer semifinal rescheduling', 'stadium venue change', 'field maintenance reason']
2026-08-31T09:32:43.123992+00:00 Starting temperature=0.7, run=16
Completed temperature=0.7, run=16, latency_ms=151557, tags=['soccer semifinal reschedule', 'stadium relocation', 'field maintenance']
2026-08-31T09:35:14.683949+00:00 Starting temperature=0.7, run=17
Completed temperature=0.7, run=17, latency_ms=106172, tags=['soccer semifinal rescheduling', 'stadium location change', 'field maintenance reason']
2026-08-31T09:37:00.857342+00:00 Starting temperature=0.7, run=18
Completed temperature=0.7, run=18, latency_ms=88382, tags=['soccer semifinal rescheduling', 'stadium venue change', 'event logistics update']
2026-08-31T09:38:29.242821+00:00 Starting temperature=0.7, run=19
Completed temperature=0.7, run=19, latency_ms=77482, tags=['soccer semifinal rescheduling', 'spartan stadium location change', 'community center backup venue']
2026-08-31T09:39:46.727933+00:00 Starting temperature=0.7, run=20
Completed temperature=0.7, run=20, latency_ms=116941, tags=['soccer semifinal rescheduling', 'spartan stadium event details', 'field maintenance impact']
```

```
2026-08-31T09:41:43.672359+00:00 Starting temperature=0.0, run=1
Completed temperature=0.0, run=1, latency_ms=78533, tags=['soccer match rescheduling', 'venue change notification', 'event logistics update']
2026-08-31T09:43:02.210598+00:00 Starting temperature=0.0, run=2
Completed temperature=0.0, run=2, latency_ms=92203, tags=['soccer match rescheduling', 'venue change', 'event logist ics']
2026-08-31T09:44:34.418499+00:00 Starting temperature=0.0, run=3
Completed temperature=0.0, run=3, latency_ms=88365, tags=['soccer match rescheduling', 'venue change', 'event logist ics']
2026-08-31T09:46:02.785399+00:00 Starting temperature=0.0, run=4
Completed temperature=0.0, run=4, latency_ms=88281, tags=['soccer match rescheduling', 'venue change', 'event logist ics']
2026-08-31T09:47:31.070511+00:00 Starting temperature=0.0, run=5
Completed temperature=0.0, run=5, latency_ms=88256, tags=['soccer match rescheduling', 'venue change', 'event logist ics']
2026-08-31T09:48:59.331103+00:00 Starting temperature=0.0, run=6
Completed temperature=0.0, run=6, latency_ms=88098, tags=['soccer match rescheduling', 'venue change', 'event logist ics']
2026-08-31T09:50:27.432519+00:00 Starting temperature=0.0, run=7
Completed temperature=0.0, run=7, latency_ms=82749, tags=['soccer match rescheduling', 'venue change', 'event logist ics']
2026-08-31T09:51:50.184615+00:00 Starting temperature=0.0, run=8
Completed temperature=0.0, run=8, latency_ms=89246, tags=['soccer match rescheduling', 'venue change', 'event logist ics']
2026-08-31T09:53:19.432699+00:00 Starting temperature=0.0, run=9
Completed temperature=0.0, run=9, latency_ms=91327, tags=['soccer match rescheduling', 'venue change', 'event logist ics']
2026-08-31T09:54:50.761252+00:00 Starting temperature=0.0, run=10
Completed temperature=0.0, run=10, latency_ms=93800, tags=['soccer match rescheduling', 'venue change', 'event logis tics']
2026-08-31T09:56:24.565470+00:00 Starting temperature=0.0, run=11
Completed temperature=0.0, run=11, latency_ms=92287, tags=['soccer match rescheduling', 'venue change', 'event logis tics']
2026-08-31T09:57:56.847973+00:00 Starting temperature=0.0, run=12
Completed temperature=0.0, run=12, latency_ms=82133, tags=['soccer match rescheduling', 'venue change', 'event logis tics']
2026-08-31T09:59:18.983479+00:00 Starting temperature=0.0, run=13
Completed temperature=0.0, run=13, latency_ms=80201, tags=['soccer match rescheduling', 'venue change', 'event logis tics']
2026-08-31T10:00:39.188906+00:00 Starting temperature=0.0, run=14
Completed temperature=0.0, run=14, latency_ms=80211, tags=['soccer match rescheduling', 'venue change', 'event logis tics']
2026-08-31T10:01:59.402639+00:00 Starting temperature=0.0, run=15
Completed temperature=0.0, run=15, latency_ms=80224, tags=['soccer match rescheduling', 'venue change', 'event logis tics']
2026-08-31T10:03:19.631694+00:00 Starting temperature=0.0, run=16
Completed temperature=0.0, run=16, latency_ms=80356, tags=['soccer match rescheduling', 'venue change', 'event logis tics']
2026-08-31T10:04:39.992330+00:00 Starting temperature=0.0, run=17
Completed temperature=0.0, run=17, latency_ms=80782, tags=['soccer match rescheduling', 'venue change', 'event logis tics']
2026-08-31T10:06:00.775789+00:00 Starting temperature=0.0, run=18
Completed temperature=0.0, run=18, latency_ms=80119, tags=['soccer match rescheduling', 'venue change', 'event logis tics']
2026-08-31T10:07:20.897509+00:00 Starting temperature=0.0, run=19
Completed temperature=0.0, run=19, latency_ms=80216, tags=['soccer match rescheduling', 'venue change', 'event logis tics']
2026-08-31T10:08:41.118909+00:00 Starting temperature=0.0, run=20
Completed temperature=0.0, run=20, latency_ms=80231, tags=['soccer match rescheduling', 'venue change', 'event logis tics']
Experiment complete: 40 total runs.
```

Metric	Temp 0.0	Temp 0.7
Distinct tag sets	2	20
Tags in all 20 runs	soccer match rescheduling	None
Tags in exactly 1 run	event logistics update; venue change notification	backup venue preparation; community center backup venue; community event logistics; community soccer semifinal rescheduling; event logistics changes; field maintenance; field maintenance notice; field maintenance update; logistical updates; logistics and backup venue; rain backup venue info; spartan stadium event details; stadium location change; stadium relocation; venue change logistics; venue change notice; venue logistics update

Latency		
Metric	Temp 0.0	Temp 0.7
p50	82,441 ms	103,358 ms
p95	92,363 ms	142,961 ms
p99	93,513 ms	149,838 ms

Part 4

Model Client and Token Accounting

1. Why is prior conversation context resent with every turn?

Prior conversation context is resent with every turn because the model does not remember earlier messages on its own. Sending the history again gives it the context needed to respond consistently to the current request.

2. How is a system prompt different from a user message?

A system prompt gives the model its overall role and rules, such as responding with only bullet code reviews. A user message is the specific request or question the user wants answered.

3. Why do input tokens grow over a conversation?

Input tokens grow because each new request includes the previous conversation history in addition to the new user message. As the conversation gets longer, there is more text for the model to read each time.

4. What eventually limits that growth?

This growth is eventually limited by the model's context window, which is the maximum number of tokens it can process at once. Once that limit is reached, older messages need to be removed or summarized.

Output:

```
(.venv) (base) annie@mac data260-2383 % python code/hw1_client.py
Homework 1 Model Client
Enter code-review requests.
For multi-line input (e.g. pasting code), type END on its own line to submit.
Commands: /stats and /exit

You: Review this Python function for correctness and edge cases:

def average(numbers):
    return sum(numbers) / len(numbers)
END

Assistant:
- **ZeroDivisionError**: Divides by `len(numbers)` which is zero if the list is empty, causing a crash.
- **Type Safety**: Assumes `numbers` contains only numeric values; non-numeric inputs will raise `TypeError`.
- **Input Validation**: Missing check for empty input list, leading to undefined behavior.
- **Edge Case Handling**: No fallback for empty lists (e.g., return 0 or raise `ValueError`).
- **Security Concern**: Unhandled division by zero could cause crashes in production code.
- **Reliability**: Lack of input validation reduces robustness for unexpected inputs.
- **Suggestion**: Add check for empty list and validate element types (e.g., `isinstance(x, (int, float))`).
Example: `if not numbers: raise ValueError("Cannot compute average of empty list")`.

--- Token Usage ---
Input tokens: 88
Output tokens: 647
Total tokens: 735
```


You: Review this Python function:

```
def is_valid_email(email):  
    return "@" in email  
END
```

Assistant:

- **Incorrect Logic**: Checks only for `@` in the string, but valid emails require more structure (e.g., domain, TLD).
- **False Positives**: Accepts invalid formats like `"test@.com"` or `"test@com"`.
- **Missing Requirements**: No validation for domain format, TLD, or proper email structure.
- **Security Concern**: Could allow malicious inputs if used for authentication without additional checks.
- **Edge Case Handling**: Fails to reject empty strings, emails with multiple `@` symbols, or missing domains.
- **Suggestion**: Use regex (e.g., `^[a-zA-Z0-9_+]+@[a-zA-Z0-9-]+\.[a-zA-Z0-9-]+$`) or a dedicated email validation library.

--- Token Usage ---

Input tokens: 283

Output tokens: 599

Total tokens: 882

You: Review this JavaScript code:

```
function saveName(name) {  
    localStorage.setItem("name", name);  
    document.getElementById("message").innerHTML = name;  
}  
END
```

Assistant:

- **Null Reference Error**: Assumes `document.getElementById("message")` exists; if missing, `innerHTML` will throw an error.
- **XSS Vulnerability**: Directly using `innerHTML` with user-provided `name` could allow HTML injection attacks.
- **Security Concern**: Lacks input sanitization or validation for the `name` parameter.
- **Suggestion**: Use `textContent` instead of `innerHTML` for safer DOM updates. Add a check for the element's existence before modifying it. Example: `const msg = document.getElementById("message"); if (msg) { msg.textContent = name; }`

--- Token Usage ---

Input tokens: 489

Output tokens: 576

Total tokens: 1065

Stats after turn 3:

You: /stats

--- Conversation Statistics ---

Turn count: 3

Cumulative input tokens: 860

Cumulative output tokens: 1822

Serialized conversation-history length: 2919 characters

You: Review this Python code:

```
def read_config(path):  
    file = open(path)  
    return file.read()  
END
```

Assistant:

- **Syntax Error**: `file.read()` returns a string, but the division operator `/` is invalid for string operations, causing a `TypeError`.
- **Resource Leak**: Opens a file but does not close it, leading to potential resource leaks.
- **No Error Handling**: Fails silently or crashes on file I/O errors (e.g., missing files, permission issues).
- **Missing Input Validation**: Does not check if `path` is valid or exists before attempting to open it.
- **Security Concern**: Exposes potential vulnerabilities if the file path is user-controlled (e.g., directory traversal attacks).
- **Suggestion**: Use `with open(path, 'r') as file:` to ensure proper file closure. Add error handling (e.g., `try/except`). Validate and sanitize the `path` input. Return `file.read()` without the division operator.

--- Token Usage ---

Input tokens: 647

Output tokens: 692

Total tokens: 1339

You: Review this JavaScript function:

```
function findUser(users, id) {  
  return users.filter(user => user.id === id)[0].name;  
}  
END
```

Assistant:

- **Undefined Result**: Returns `undefined` if no user matches the `id`, leading to potential errors when using the result.
- **No Input Validation**: Assumes `users` is an array and `id` is a valid type; invalid inputs may cause runtime errors.
- **Incorrect Method Use**: Uses `filter` and `[0]` instead of `find` for direct lookup, which is less efficient and error-prone.
- **Security Concern**: No validation for user object structure (e.g., missing `id` or `name` fields).
- **Suggestion**: Replace with `find` and add a null check: `const user = users.find(u => u.id === id); return user?.name;`. Validate `users` is an array and handle missing/invalid `id`.

--- Token Usage ---

Input tokens: 865

Output tokens: 739

Total tokens: 1604

Stats after turn 5:

You: /stats

--- Conversation Statistics ---

Turn count: 5

Cumulative input tokens: 2372

Cumulative output tokens: 3253

Serialized conversation-history length: 4768 characters

Final cumulative stats:

--- Final Cumulative Statistics ---

Turn count: 5

Cumulative input tokens: 2372

Cumulative output tokens: 3253